Independent Products in Real L_1: Full-Excess Bounds, Gamma Multipliers, and an Integrable Renorming Obstruction

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M642

October 9, 2026

Abstract

We examine the actual first-exit laws in the source construction of independent path products in real L_1. Keeping the full Gaussian excess after conditional symmetrization, and applying a one-sided variance bound before a joint-convexity argument, gives midpoint estimates that dominate all parameter choices in the source's exponential and Gaussian-square bounds. The same calculation extends the construction to every mean-one Gamma multiplier with positive shape. Separately, a density-weighted cylinder approximation removes the second-moment assumption from the bounded-path obstruction: every nonnegative, nonconstant, integrable mean-one multiplier yields a product span with no equivalent asymptotically uniformly convex norm. Positive midpoint bounds are not asserted for this entire integrable class. A sign-split integral certificate evaluates the Gaussian excess without Taylor expansion across the absolute-value cusp, with explicit panel and infinite-tail errors. Classical mixture, symmetrization, Cantelli, and convexity arguments are credited; no optimal modulus or historical priority is claimed. Contribution assessment: C3/5, a source-specific full-excess improvement and Gamma-law extension together with a proved first-moment weakening of the renorming obstruction.

Contents

1 Source scope and mathematical setting **2**

2 Conditional laws and their physical first moments **3**

3 Full Gaussian excess and Gamma multipliers **3**

4 Comparison with all source parameter choices **5**

5 Removing the second moment from the obstruction **7**

6 A sign-split finite integral certificate **8**

7 Advantages, costs, and remaining questions **10**

8 Evidence and contribution boundary **10**

1 Source scope and mathematical setting

The assigned September 27, 2026 manuscript [1] treats the countably branching tree $\mathcal{T} = \mathbb{N}^{<\omega}$. On the canonical product probability space take independent copies W_v of a real nonnegative mean-one random variable, indexed by nonroot vertices. Writing v^- for the parent, set

$$P_\varnothing = 1, \qquad P_v = P_{v^-} W_v, \qquad X_W = \overline{\operatorname{span}_{\mathbb{R}}\{P_v : v \in \mathcal{T}\}}^{L_1}. \tag{1}$$

Every norm in the product space below is the inherited $\|f\|_1 = \mathbb{E}|f|$, not a Euclidean norm on coefficients. Positivity and independence give $\|P_v\|_1 = 1$. The source assumes $W > 0$, nonconstant, $\mathbb{E}W = 1$, and $m_2 = \mathbb{E}W^2 < \infty$. It proves no AUC renorming under these hypotheses and positive averaged midpoint bounds for W exponential or $W = G^2$, G standard real Gaussian.

For a norm N, let $\operatorname{cof}(X)$ be its closed finite-codimensional subspaces, and define

$$\widehat{\delta}_N(t) = \inf_{N(x)=1} \sup_{F \in \operatorname{cof}(X)} \inf_{\substack{y \in F \\ N(y) \geq 1}} \left(\frac{N(x+ty) + N(x-ty)}{2} - 1 \right), \tag{2}$$

$$\overline{\delta}_N(t) = \inf_{N(x)=1} \sup_{F \in \operatorname{cof}(X)} \inf_{\substack{y \in F \\ N(y)=1}} (N(x+ty) - 1). \tag{3}$$

The head subspace must be chosen before the direction. The standard AMUC modulus uses the maximum of the two endpoint norms; positivity of (2) implies AMUC, but numerical constants for these two definitions must not be silently identified.

Source module	Mechanism and treatment in this report
Introduction	Product probability space, actual L_1 norm, averaged modulus, and separate distributional hypotheses.
Conditional kernels	Finite-rank conditional expectation on the closed product span; independent first exits; two first-moment estimates. Reproduced in Section 2.
Exponential products	Laplace/normal mixture and an affine excess bound. Its valid bounds are sharpened, not declared erroneous.
Gaussian squares	Orthogonal polarization, a probability route paying a union-bound factor, and an even-convex difference route. Both remain valid and are compared with the new envelope.
Obstruction	Bounded norm-one paths and weakly null child increments. Its L_2 proof is replaced by an L_1 proof, and the actual cofinite branch argument is retained.

This construction is not the segment-space norm in M638 or the bounded-potential norm of its companion. Their averaged coefficient $1/4$ is not transferred here. The source explicitly makes no assertion of reflexivity, the Daugavet property, or precompactness in measure of the unit ball; neither do we.

The current public context includes Baudier's September 18, 2026 preprint [3], which establishes AMUC for the Kadets–Werner modification of Bourgain–Rosenthal space and states failure of AUC renormability. That is a different construction. The Kadets–Werner primary abstract [4] and current versions of the source's probability references were freshly checked. The general separation of AMUC and AUC renormability is not new here. Normal/Gamma mixtures are classical [7, 8]; the contribution is their use in this source's conditional, all-directions argument. The public search did not supply a prior theorem for precisely the Gamma product spans below, but a bounded search cannot establish historical novelty.

2 Conditional laws and their physical first moments

Let T be a finite predecessor-closed subtree containing the root, and let $\mathcal{G}_T$ be generated by its nonroot multipliers. For the last ancestor v_T of v in T, independence gives

$$Q_T P_v = \mathbb{E}(P_v \mid \mathcal{G}_T) = P_{v_T}. \tag{4}$$

Thus Q_T is a contractive projection on X_W with finite range $E_T = \operatorname{span}\{P_v : v \in T\}$; $F_T = \ker Q_T$ is closed and finite-codimensional. Continuity and finite-dimensional closedness prove this on the completion. Applying $I - Q_T$, whose norm is at most two, to finite-product approximations proves that finite-product elements of F_T are dense in F_T. We do not claim $I - Q_T$ is contractive.

For a finite-product tail $y \in F_T$, group terms by their first vertices i outside T. The source's exact factorization is

$$y = d + \sum_i W_i B_i, \qquad d = -\sum_i m_i, \quad m_i = \mathbb{E}(B_i \mid \mathcal{G}_T), \quad \sigma = \Big(\sum_i B_i^2\Big)^{1/2}. \tag{5}$$

The coefficients depend on old variables and strict descendants of their exits, never on the W_i at the exits. Given $\mathcal{G}_T$, the B_i are independent. Given $\mathcal{H}_T = \mathcal{G}_T \vee \sigma(B_i : i)$, the W_i remain independent with their original law. These are different conditioning operations; resampling the coefficients together with the exits would change the scalar calculation.

Lemma 2.1 (Source first-moment constants retained)**.** *If $m_2 < \infty$, then, both conditionally on $\mathcal{G}_T$ and after averaging,*

$$\|y\|_1 \le C_W \mathbb{E}\sigma, \qquad C_W = \min\{\, 2 + \sqrt{m_2 - 1},\, 2\sqrt{m_2}\,\}. \tag{6}$$

Proof. Write $y = \sum B_i(W_i - 1) + \sum(B_i - m_i)$. After fixing the coefficients the first sum has variance $(m_2 - 1)\sigma^2$; hence its expected absolute value is bounded by $\sqrt{m_2 - 1}\mathbb{E}\sigma$. Conditional independent copies B_i' and independent signs ε_i give

$$\mathbb{E}\Big|\sum(B_i - m_i)\Big| \le \mathbb{E}\Big|\sum(B_i - B_i')\Big| = \mathbb{E}\mathbb{E}_\varepsilon\Big|\sum \varepsilon_i(B_i - B_i')\Big| \le 2\mathbb{E}\sigma.$$

Alternatively $U_i = W_i B_i$ are conditionally independent and $y = \sum(U_i - \mathbb{E}(U_i \mid \mathcal{G}_T))$. Symmetrization and the triangle inequality bound $\mathbb{E}|y|$ by $2\mathbb{E}(\sum W_i^2 B_i^2)^{1/2} \le 2\sqrt{m_2}\mathbb{E}\sigma$. Every expectation here is first taken under the declared conditional law. Integrate over the old coordinates to conclude. These are classical symmetrization steps [6], already proved in the source; selecting their minimum is not itself new. □

For real scalars, $(|a + b| + |a - b|)/2 = \max\{|a|, |b|\}$. Consequently

$$M_t(u, y) := \frac{\|u + ty\|_1 + \|u - ty\|_1}{2} - \|u\|_1 = \mathbb{E}(t|y| - |u|)_+. \tag{7}$$

This identity fails in this form for complex scalars. It is the reason that the absolute-value cusp must be respected rather than replaced by a quadratic physical metric.

3 Full Gaussian excess and Gamma multipliers

Let G be standard Gaussian, and for $s, b \ge 0$ put

$$H(s, b) = \mathbb{E}(s|G| - b)_+, \qquad h(z) = \mathbb{E}(|G| - z)_+, \quad H(s, b) = sh(b/s) \quad (s > 0). \tag{8}$$

Define $H(0,b)=0$. The expectation of the positive part of an affine function is jointly convex; H is increasing in s and decreasing in b. It is finite, continuous, and strictly positive when $s>0$ and $b<\infty$.

Lemma 3.1 (One-sided variance estimate)**.** *If $\mathbb{E}V=1$, $\operatorname{Var}V\le v$, and $0<\theta<1$, then*

$$\mathbb{P}(V\ge\theta)\ge q_v(\theta):=\frac{(1-\theta)^2}{v+(1-\theta)^2}. \tag{9}$$

Proof. For $Z=1-V$, $d=1-\theta>0$, and $c\ge 0$, Markov's inequality on $\{Z\ge d\}$ gives $\mathbb{P}(Z\ge d)\le (v+c^2)/(d+c)^2$. Take $c=v/d$. The resulting bound $v/(v+d^2)$ also bounds $\mathbb{P}(V<\theta)$; complementing proves the assertion. The zero-variance case is immediate. This is Cantelli's classical inequality, not a new concentration principle. □

Lemma 3.2 (The actual Gamma difference mixture)**.** *Let W have Gamma shape $a>0$ and rate a, with density*

$$\frac{a^a}{\Gamma(a)}w^{a-1}e^{-aw}\quad (w>0).$$

For independent copies W,W', a Gamma variable U of shape a, rate one, and an independent standard Gaussian G,

$$W-W'\ \overset{d}{=}\ \frac{\sqrt{2U}}{a}G. \tag{10}$$

For fixed real B_i, $\sigma>0$, the half-difference of the conditionally copied variables in (5) *has the law*

$$\frac{y-y'}{2}\ \overset{d}{=}\ \sigma\sqrt{\frac{V}{2a}}G,\qquad V=\sum_i\frac{B_i^2}{\sigma^2}\frac{U_i}{a},\quad \mathbb{E}V=1,\quad \operatorname{Var}V=\frac{\kappa}{a},\quad \kappa=\frac{\sum_i B_i^4}{\sigma^4}\le 1. \tag{11}$$

Proof. The Gamma integral gives $\mathbb{E}e^{iqW}=(1-iq/a)^{-a}$, using the branch continuous at zero. The characteristic function of $W-W'$ is $(1+q^2/a^2)^{-a}$. Conditional Gaussian integration on the right of (10) gives $\mathbb{E}e^{-q^2U/a^2}=(1+q^2/a^2)^{-a}$, proving the identity. Use independent copies of this representation for all exits. Conditional on the U_i, their sum is Gaussian with the variance in (11). The Gamma moments $\mathbb{E}U_i=a$, $\operatorname{Var}U_i=a$ and independence give the last assertions. This is the classical symmetric variance-Gamma mixture [8], with its normalization verified here. □

Theorem 3.3 (Full-excess midpoint bound for every Gamma shape)**.** *Let X_a be the product span* (1) *for the preceding Gamma law. Set*

$$C_a=\min\{2+a^{-1/2},\,2\sqrt{1+a^{-1}}\},\quad \alpha_a(\theta)=\frac{\sqrt{\theta}}{\sqrt{2a}\,C_a}. \tag{12}$$

For every $t>0$ and $0<\theta<1$,

$$M_t(u,y)\ge q_{1/a}(\theta)H\left(t\alpha_a(\theta)\|y\|_1,\|u\|_1\right)\quad (u\in E_T,\ y\in F_T), \tag{13}$$

$$\widehat{\delta}_{X_a}(t)\ge L_a(t):=\sup_{0<\theta<1}q_{1/a}(\theta)H\left(t\alpha_a(\theta),1\right)>0. \tag{14}$$

The norm and conditional law are the actual ones in (1).

Proof. First take finite-product y. Given $\mathcal{H}_T$, the function $\phi(z) = (t|z| - |u|)_+$ is even and convex. With a conditional independent copy y' using only fresh exits,

$$\mathbb{E}(\phi(y) \mid \mathcal{H}_T) \geq \mathbb{E}\left(\phi((y - y')/2) \mid \mathcal{H}_T\right).$$

Lemma 3.2, monotonicity in the Gaussian scale, and Lemma 3.1 imply the sharper conditional assertion

$$\mathbb{E}(\phi(y) \mid \mathcal{H}_T) \geq q_{\kappa/a}(\theta) H\left(t\sqrt{\theta/(2a)}\,\sigma, |u|\right). \tag{15}$$

When $\sigma = 0$, the right side is set to zero and no normalization is used. To obtain a bound uniform over all coefficient lists, replace κ by one. Joint convexity of H, now applied to the old data and the coefficient laws, gives

$$M_t(u, y) \geq q_{1/a}(\theta) H\left(t\sqrt{\theta/(2a)}\,\mathbb{E}\sigma, \|u\|_1\right).$$

Since $m_2 = 1 + 1/a$, Lemma 2.1 and monotonicity in the first argument give (13).

For a completed tail, use density of finite-product tails and

$$|M_t(u, y) - M_t(u', y')| \leq 2\|u - u'\|_1 + t\|y - y'\|_1. \tag{16}$$

The right side of (13) is continuous in the two norms, so the bound passes to every $y \in F_T$; no infinite first-exit expansion is asserted. For a unit center x, choose a finite-product u with $\|x - u\|_1 < \varepsilon$, then choose T containing all its support paths. For all $y \in F_T$ with $\|y\|_1 \geq 1$,

$$M_t(x, y) \geq q_{1/a}(\theta) H(t\alpha_a(\theta), 1 + \varepsilon) - 2\varepsilon.$$

Take the inner infimum, the supremum over cofinite spaces, then $\varepsilon \downarrow 0$, and finally the infimum over centers. This proves the bound for each θ, hence for their supremum. Strict positivity follows for any fixed $0 < \theta < 1$. □

The proof extends the two source laws: $a = 1$ is exponential and $a = 1/2$ is G^2. It does not assert AMUC for arbitrary integrable multipliers. The representation-sensitive κ/a in (15) may improve a particular finite law, but cannot be replaced by a favorable deterministic number in (14) without a uniform hypothesis over every direction in the chosen cofinite subspace.

4 Comparison with all source parameter choices

At $\theta = 1/2$, Theorem 3.3 gives the explicit bounds

$$B_{\exp}(t) = \frac{1}{5} H\left(\frac{t}{4\sqrt{2}}, 1\right), \qquad B_{\mathrm{sq}}(t) = \frac{1}{9} H\left(\frac{t}{\sqrt{2}(2 + \sqrt{2})}, 1\right). \tag{17}$$

The constants $1/5$ and $1/9$ use the variance, rather than the source's raw second-moment lower-probability estimates. Keeping the full positive-part expectation avoids a separate threshold event on the old coordinates.

Proposition 4.1 (Uniform domination of the source envelopes)**.** *Let $S_{\exp}(t,K)$, $S_{24}(t,K)$, and $S_{12}(t,K)$ denote, respectively, the source's three parameterized lower bounds:*

$$\begin{aligned} S_{\exp}(t,K) &= \frac{h(2K)}{16}\left(\frac{t}{3}-\frac{1}{K}\right)_+, \\ S_{24}(t,K) &= \frac{\mathbb{P}(|G|\ge\sqrt{2}K)}{24}\left(\frac{tK}{2+\sqrt{2}}-1\right)_+, \\ S_{12}(t,K) &= \frac{\mathbb{P}(|G|\ge\sqrt{2}K)}{12}\left(\frac{tK}{2\sqrt{3}}-1\right)_+. \end{aligned} \tag{18}$$

For every $t,K>0$,

$$B_{\exp}(t)\ge\frac{8}{5}S_{\exp}(t,K), \qquad B_{\mathrm{sq}}(t)\ge\frac{8}{3}S_{24}(t,K), \qquad B_{\mathrm{sq}}(t)\ge\frac{4}{3}S_{12}(t,K). \tag{19}$$

The inequalities therefore also hold after taking the supremum over all source parameters K.

Proof. For $s,z\ge0$,

$$H(s,1)\ge(sz-1)_+\mathbb{P}(|G|\ge z).$$

Set $s=t/(\sqrt{2}(2+\sqrt{2}))$, $z=\sqrt{2}K$. This proves the 24-bound comparison. Since $2+\sqrt{2}<2\sqrt{3}$, it also proves the 12-bound comparison.

For exponentials, replacing $C_1=2\sqrt{2}$ by the weaker source constant three decreases the new bound to

$$\frac{1}{5}H(t/6,1)=\frac{t}{30}h(6/t).$$

If $K\le3/t$, the old bound is zero. Otherwise $2K>6/t$, and h is decreasing, while

$$S_{\exp}(t,K)=\frac{t}{48}h(2K)\left(1-\frac{3}{tK}\right)\le\frac{t}{48}h(6/t).$$

The ratio $48/30=8/5$ proves the result. □

These comparisons concern certified lower envelopes for the same inherited norm. They do not identify the actual asymptotic modulus. In particular a larger lower bound is not a measured algorithmic speedup or a new separation of Banach-space properties.

Proposition 4.2 (Small-scale behavior of the new envelope)**.** *For each fixed $a>0$, the lower envelope L_a in* (14) *satisfies*

$$\lim_{t\downarrow0}t^2\log L_a(t)=-aC_a^2. \tag{20}$$

Thus the constants are 8 *for exponentials and $(2+\sqrt{2})^2/2$ for Gaussian squares. This is an asymptotic of the proved bound, not of $\widehat{\delta}_{X_a}$.*

Proof. For $z\ge0$, direct translation in the Gaussian integral gives

$$h(z)=\sqrt{\frac{2}{\pi}}e^{-z^2/2}\int_0^\infty se^{-zs-s^2/2}\,ds. \tag{21}$$

For $z\ge1$, the integral is at most z^{-2}. Restricting to $0\le s\le1/z$ bounds it below by $e^{-3/2}/(2z^2)$. Hence $\log h(z)=-z^2/2-2\log z+O(1)$. Because $q_{1/a}(\theta)\le1$ and $\alpha_a(\theta)\le1/(\sqrt{2a}C_a)$,

$$L_a(t)\le H(t/(\sqrt{2a}C_a),1).$$

For the matching lower logarithmic bound, choose $\theta=1-t^2$ for $0<t<1$. Then $\log q_{1/a}(\theta)=4\log t+O(1)$ and $\theta\to1$. Insert the preceding asymptotic into this single admissible term of (14). Multiplication by t^2 eliminates all logarithmic prefactors and gives (20). □

5 Removing the second moment from the obstruction

The next result concerns a different set of hypotheses from the Gamma midpoint estimate. Now assume only

$$W \geq 0, \qquad \mathbb{E}W = 1, \qquad W \text{ is not almost surely constant.} \tag{22}$$

Finite products remain integrable with norm one. Infinite dimensionality does not need L_2 orthogonality: if $\sum_{j=1}^{k} c_j(W_j - 1) = 0$, conditioning on W_i gives $c_i(W_i - 1) = 0$, hence $c_i = 0$. This proves independence of the centered first-generation variables in L_1.

Lemma 5.1 (Weak nullity using only the first moment)**.** *For each fixed vertex v, the child increments*

$$D_{v,n} = P_v(W_{v^\frown n} - 1)$$

are weakly null in X_W, and $\|D_{v,n}\|_1 = \rho := \mathbb{E}|W - 1| > 0$.

Proof. Define the probability measure $d\nu_v = P_v\, d\mathbb{P}$. This is legitimate even if P_v vanishes on a set of positive measure: nonnegativity and $\mathbb{E}P_v = 1$ suffice. Its density involves only the finite ancestor path. The child multipliers therefore remain independent with their original laws under ν_v, independent of all other coordinates.

For any bounded measurable g, choose bounded cylinder approximations g_m, depending on finitely many coordinates, such that $\|g_m\|_\infty \leq \|g\|_\infty$ and $\int |g - g_m|\, d\nu_v \to 0$. Such approximations follow by conditional expectations on increasing finite-coordinate sigma fields; their union generates the canonical countable product sigma field. Put $\xi_n = W_{v^\frown n} - 1$. For all sufficiently large n, the cylinder omits that child, so $\int \xi_n g_m\, d\nu_v = 0$. For every $K > 0$,

$$\left| \int \xi_n (g - g_m)\, d\nu_v \right| \leq K \int |g - g_m|\, d\nu_v + 2\|g\|_\infty \mathbb{E}\big[|W - 1| 1_{\{|W-1|>K\}}\big]. \tag{23}$$

First choose K using integrability, then m using cylinder density, then n outside the finite cylinder. This proves $\int g D_{v,n}\, d\mathbb{P} \to 0$ for every $g \in L_\infty(\mathbb{P})$. Every bounded functional on the closed subspace X_W extends to $L_1(\mathbb{P})$ by Hahn–Banach and is represented by an L_∞ test. Thus the sequence is weakly null in X_W. The norm identity follows from independence and $\mathbb{E}P_v = 1$. □

Theorem 5.2 (Integrable product obstruction)**.** *Under* (22)*, no equivalent norm on X_W is AUC. More precisely, if*

$$A_0\|z\|_1 \leq N(z) \leq B_0\|z\|_1, \qquad 0 < A_0 \leq B_0 < \infty,$$

then

$$\overline{\delta}_N\left(\frac{A_0\rho}{2B_0}\right) = 0. \tag{24}$$

Proof. The modulus is nonnegative: at each unit center a norming functional supplies a cofinite kernel where the norm is at least one. Suppose its value at $t_0 = A_0\rho/(2B_0)$ were $\delta > 0$, and put $\gamma = \delta/2$. At a vertex v, choose the cofinite space F for the center $P_v/N(P_v)$. The definition gives

$$N(P_v + z) \geq (1 + \gamma)N(P_v) \quad \text{if } z \in F, \quad N(z) \geq t_0 N(P_v).$$

The extension from equality to larger radii follows from convexity: $(N(x+sw)-1)/s$ is nondecreasing in $s > 0$ when $N(x) = N(w) = 1$.

By Lemma 5.1, the quotient images of $D_{v,n}$ in the finite-dimensional space X_W/F tend to zero in norm. Choose $z_n \in F$ with $N(D_{v,n} - z_n) \to 0$. Eventually $N(z_n) \geq A_0\rho/2 \geq t_0 N(P_v)$. Selecting n so that the approximation error is at most $\gamma N(P_v)/2$ gives

$$N(P_{v^\frown n}) \geq (1 + \gamma/2)N(P_v).$$

Iterate, choosing a new cofinite space and child at each vertex. The same γ works at every depth, so the resulting depth-k product has norm at least $A_0(1+\delta/4)^k$, contrary to the upper bound $B_0\|P_v\|_1 = B_0$. Finally $0 < \rho < 2$, since $\rho = 2 - 2\mathbb{E}\min\{W, 1\}$, and the latter expectation is positive for any nonnegative mean-one law. Thus $0 < t_0 < 1$. □

This preserves the source's branch selection argument and weak topology while replacing only the finite-L_2 step. It also permits an atom at zero. The result is a proved assumption extension of this construction, not a claim about every subspace of L_1. Combining Theorems 3.3 and 5.2, each X_a, $a > 0$, is AMUC in its inherited norm and has no equivalent AUC norm.

Example 5.3 (Infinite variance is genuinely allowed)**.** Take $1 < \beta < 2$, $w_0 = (\beta - 1)/\beta$, and $\mathbb{P}(W > s) = (w_0/s)^\beta$ for $s \geq w_0$. Then $\mathbb{E}W = \beta w_0/(\beta - 1) = 1$, while $\mathbb{E}W^2 = \infty$. Moreover

$$\rho = 2\mathbb{E}(W-1)_+ = \frac{2w_0^\beta}{\beta - 1}.$$

Theorem 5.2 applies, although the source's L_2-orthogonality proof does not. No positive midpoint bound for this Pareto product span is asserted.

Proposition 5.4 (A necessary distinction between the conclusions)**.** *Let $W = 1 + \varepsilon R$, where R is a symmetric sign and $0 < \varepsilon \leq 1$. Then the inherited product norm satisfies $\widehat{\delta}_{X_W}(t) = 0$ for $0 < t \leq 1$, although it has no equivalent AUC norm.*

Proof. The first-generation signs $R_n = (W_n - 1)/\varepsilon$ belong to X_W, have norm one, and are weakly null by Lemma 5.1 at the root. For any cofinite F, choose $z_n \in F$ with $\|z_n - R_n\|_1 \to 0$, and normalize $w_n = z_n/\|z_n\|_1$. Then $\|w_n - R_n\|_1 \to 0$. At the unit center 1,

$$M_t(1, R_n) = \mathbb{E}(t|R_n| - 1)_+ = (t-1)_+ = 0 \quad (0 < t \leq 1).$$

Continuity gives $\inf_{w \in F, \|w\|_1 \geq 1} M_t(1, w) = 0$. Every such F has this property, so the supremum and then the outer infimum give zero. Nonnegativity supplies the reverse inequality. The no-AUC assertion follows from Theorem 5.2. □

6 A sign-split finite integral certificate

The original Ram local-moment chapter [2] proves a Taylor formula with integral remainder under derivatives through each integration segment. Those assumptions do not hold across the cusp of $(t|z| - b)_+$. We first split at the threshold, and only then apply the smooth formula to the Gaussian density. This use of the original proof retains its regularity and remainder hypotheses; positivity of a finite moment matrix is not substituted for a law.

For $z \geq 0$, the exact translated integral is

$$h(z) = \sqrt{2/\pi}\, e^{-z^2/2} I(z), \qquad I(z) = \int_0^\infty s e^{-zs - s^2/2}\, ds. \tag{25}$$

Unlike subtracting two almost equal Gaussian-tail expressions, this representation has a nonnegative integrand. The following statement is a finite, rationally executable certificate, not a speed claim.

Proposition 6.1 (Exact polynomial panels and discarded tail)**.** *Let $z \geq 0$, $S > 0$, and $n \geq 0$ be an integer. Define*

$$J_n(z, S) = \sum_{j=0}^{n} \frac{(-1)^j}{j!} \sum_{m=0}^{j} \binom{j}{m} \frac{z^{j-m}}{2^m} \frac{S^{j+m+2}}{j+m+2}. \tag{26}$$

For every integer $N \geq 0$,

$$\max\{0, J_{2N+1}(z,S)\} \leq I(z) \leq J_{2N}(z,S) + e^{-zS-S^2/2}. \tag{27}$$

All polynomial terms are rational for rational z, S. Certified intervals for the positive prefactor in (25) turn these inequalities into certified intervals for h and the bounds in Theorem 3.3 at any specified parameter value.

Proof. Taylor's integral remainder gives, for every $u \geq 0$,

$$\sum_{j=0}^{2N+1} \frac{(-u)^j}{j!} \leq e^{-u} \leq \sum_{j=0}^{2N} \frac{(-u)^j}{j!}.$$

No decreasing-term assumption is needed: the derivative remainder has the sign $(-1)^{n+1}$. Substitute $u = zs + s^2/2$, multiply by s, and integrate from zero to S. The binomial theorem gives (26). For the omitted interval,

$$0 \leq \int_S^\infty s e^{-zs-s^2/2}\, ds \leq e^{-zS} \int_S^\infty s e^{-s^2/2}\, ds = e^{-zS-S^2/2}.$$

The integral is nonnegative, so its lower bound may be clipped at zero. Multiplication by a positive prefactor preserves the bounds. □

At large $zS + S^2/2$, a single Taylor panel can require many terms and large rational numerators. Subdivision, or the translated large-z bounds used in Proposition 4.2, can be preferable. This is a choice of certified evaluation method, not an alteration of the distribution. Evaluating a finite list of θ's gives a valid lower bound for the supremum in (14); it does not prove that an optimizer was found.

Corollary 6.2 (Actual-law approximation budget)**.** *Suppose finite products u, y and actual L_1 elements x, z satisfy $\|x-u\|_1 \leq \epsilon_x$, $\|z-y\|_1 \leq \epsilon_y$. If a certified computation gives $M_t(u,y) \geq L$, then*

$$M_t(x,z) \geq L - 2\epsilon_x - t\epsilon_y.$$

Proof. Use (16). These are errors in the original product measure, not coefficient errors or errors under a renormalized truncation of that measure. □

For example, replacing a multiplier distribution by a bounded panel law requires its own coupling or integral error certificate. Matching some moments alone does not justify Corollary 6.2. This also separates a finite computation from the infinite cofinite-subspace quantifier in (2).

Example 6.3 (One genuine exponential exit)**.** At the root let $u = 1$, $y = W - 1$, and W be mean-one exponential. Then y is a genuine first-exit tail, $\mathbb{E}y = 0$, $\|y\|_1 = 2/e$, and its conditional coefficient scale is exactly $\sigma = 1$. Direct integration at the points $1 \pm 1/t$ yields

$$M_t(1, W-1) = \begin{cases} te^{-1-1/t}, & 0 < t \leq 1, \\ t\big(e^{-1-1/t} + e^{-1+1/t}\big) - 1, & t \geq 1. \end{cases} \tag{28}$$

Indeed the upper tail contributes $te^{-1-1/t}$. When $t > 1$, the lower interval $0 \leq W < 1 - 1/t$ contributes $te^{-1+1/t} - 1$; when $t \leq 1$ it is absent. Both formulas agree at $t = 1$. Keeping σ before the first-moment relaxation in (15) gives the valid lower bound

$$M_t(1, W-1) \geq \tfrac{1}{5} H(t/2, 1) = \frac{t}{10} h(2/t).$$

This comparison is between quantities under the actual exponential law. The tail has norm $2/e < 1$, so this particular unscaled example is not itself a direction in the unit-tail infimum of (2). Rescaling it would be legitimate, with the corresponding change in t.

7 Advantages, costs, and remaining questions

The full-excess calculation avoids two losses in the special-law source estimates: replacing a Gaussian excess by a single threshold event, and retaining an unnecessarily large first-moment constant. Cantelli further uses the variance of the normalized random scale. The resulting inequalities dominate the *optimized source parameter families*, not just their advertised choices. The Gamma calculation supplies a continuous family containing both source laws. Its proof uses classical normal mixtures, not a new mixture principle [7, 8].

The benefits have precise limits. The coefficient concentration κ is representation-dependent; a uniform Banach-space conclusion uses $\kappa \leq 1$ unless an admissible representation with a sharper bound is available for every direction. Cantelli does not exploit the full Gamma convolution law, and the constant C_a can be loose. Neither L_a nor its logarithmic scale is claimed to be the optimal midpoint modulus. A small shape can make the quantitative lower bound extremely small despite strict positivity. The integral certificate has an explicit tail and can be rigorous, but its rational arithmetic cost has not been benchmarked.

The first-moment obstruction is independent of the Gaussian-mixture argument. Its proof allows infinite variance, but gives no uniform rate of weak convergence: the finite cylinder depends on the functional and the accuracy, and the weighted measure depends on the chosen parent. It proves the needed existence of a child at each stage, not an effective branch-search complexity bound. The bounded-sign example shows why positivity of the AMUC modulus cannot be deduced from this obstruction alone.

Useful extensions would retain the exact Gamma scale law rather than Cantelli, or certify a representation-sensitive distribution of κ before taking the cofinite infimum. A different direction is to characterize those integrable multipliers whose symmetrized law has enough unbounded conditional excess for AMUC. Any such extension must survive the bounded-sign counterexample, retain control of $\mathbb{E}\sigma$ by the actual L_1 norm, and respect the order of the head and direction quantifiers. No assertion about complex L_1, arbitrary dependencies, signed mean-one multipliers, or arbitrary subspaces of L_1 follows from this report.

8 Evidence and contribution boundary

All nine cached public text files were freshly displayed in bounded, untruncated chunks, including every scientific module, bibliography, and README: 846 lines in total. SHA-256 values were persisted before display and verified afterward; there are no unread intervals in these files. Captured-tree authentication establishes byte identity, not a live-branch assertion or proof certification. The principal private original read was the 446-line main file of the local-moment chapter [2]. Its included appendix, study, and lab modules were not read and are not counted. No whole-corpus claim is made.

The fresh public context check used primary abstracts and metadata for [3, 4, 5, 7, 8], and the independent-copy/sign proof passage of [6]. Those access scopes do not constitute complete external proof audits. Search results do not establish historical priority for the Gamma family or the integrability weakening. The proofs above establish the stated results directly; the source's

valid special-law and renorming mechanisms are retained and credited. No source mathematical correction was required.

The owned executable checks test exact Gamma mixture moments, conditional scale variances, actual finite product-law norms and conditioning, and rational scalar enclosures. Separate numerical integrations are diagnostics only. Counts and full file hashes are recorded in the accompanying evidence receipts; parent and previous report cases are not included. Such checks cannot certify weak limits, all directions of a Banach space, or an optimal modulus.

The C3 assessment concerns the combination of a uniform improvement over the source's full parameter families, the fully proved Gamma extension, and the first-moment obstruction beyond the source's second-moment assumption. It is not credit for classical Taylor, Cantelli, Jensen, symmetrization, or Hahn–Banach alone. Applicability L3 describes relevance and is independent of this contribution score.

References

[1] OpenAI mathematics manuscript, *Independent products in real L_1: asymptotic midpoint convexity without AUC renormings*, September 27, 2026. Pinned local source: main file, first-exit notes, and sections on conditional kernels, exponential products, Gaussian squares, and the renorming obstruction; captured-tree provenance in the audit.

[2] M. Subbarao, *Ram Calculus*, Volume 1, revision 2, Chapter 3, original local-moment manuscript `ch03-local-moments-rev2.tex`, private local corpus. Main file lines 1–446 freshly read; included modules unread. Used for regularity-aware integral Taylor remainders and tail accounting, not for a global Banach-space theorem.

[3] F. Baudier, *The Kadets–Werner modification of Bourgain–Rosenthal space is asymptotically midpoint uniformly convex*, 2026 preprint, arXiv:2609.21283. Primary abstract/version scope; the assigned product construction is distinct.

[4] V. Kadets and D. Werner, *A Banach space with the Schur and the Daugavet property*, Proc. Amer. Math. Soc. 132 (2004), 1765–1773, arXiv:math/0301182.

[5] E. Levina and R. Vershynin, *Partial estimation of covariance matrices*, Probab. Theory Related Fields 153 (2012), 405–419, arXiv:1008.1716.

[6] P. L. Bartlett and S. Mendelson, *Rademacher and Gaussian complexities: risk bounds and structural results*, J. Mach. Learn. Res. 3 (2002), 463–482. Primary paper, Theorem 8 and independent-copy/sign proof passage.

[7] D. F. Andrews and C. L. Mallows, *Scale mixtures of normal distributions*, J. R. Stat. Soc. B 36 (1974), 99–102, publisher DOI.

[8] A. Fischer, R. E. Gaunt, and A. Sarantsev, *The variance-gamma distribution: a review*, 2023 preprint, arXiv:2303.05615. Primary abstract/version scope; the particular characteristic-function identity needed here is proved in Lemma 3.2.